



Internship Report

Name of Company

Student Name

Student card number: 00000000

Under the supervision of

Name of DEPA academic supervisor
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Name of Company supervisor
Company supervisor

Academic Year 2025–2026



I Executive Summary

[Summarise the whole report: state all the major points of your project, and outline the report's scope, purpose and conclusions.]





I Acknowledgements

[Optional. If used, it must come after the Executive Summary and before the Table of Contents.]



I Table of Contents

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I Introduction

[Briefly outline the company and/or department you worked for. Summarise the work you have done and clearly state the objectives of your project. You can also state the importance of your project and its relevance.]

I Datasets

[Optional short introduction: why these two datasets were used together, e.g. ATM'22 for the variety of sites and scanners, AIB23 for fibrotic lungs.]

II The ATM'22 dataset

The Multi-site, Multi-domain Airway Tree Modeling dataset (ATM'22) was released as part of an official challenge held in conjunction with MICCAI 2022 (Zhang et al., 2023). It targets the binary segmentation of the pulmonary airway tree from chest CT, and at the time of its release it was the largest publicly available airway dataset with complete voxel-wise annotations. The reference benchmark that preceded it, EXACT'09, provided only 40 scans and never released its manual annotations, which had long limited the training and fair evaluation of data-driven methods.

The dataset contains 500 chest CT scans, partitioned into 300 training, 50 validation and 150 test cases. The test set is held privately by the organizers and is accessible only through the submission of a Docker container, which guarantees the fairness and reproducibility of the reported results. The scans originate from three sources, as summarized in Table 1.

Table 1: Composition of the ATM'22 dataset by split and by source site.

Split	Shanghai Chest Hospital	LIDC-IDRI	EXACT'09	Total
Training	226	54	20	300
Validation	20	30	–	50
Test	84 (58 COVID-19)	66	–	150
Total	330	150	20	500

For the 20 cases taken from EXACT'09, only the airway annotations are distributed; the corresponding CT volumes must be obtained separately from the original source.

The acquisition parameters are heterogeneous, which is the basis of the multi-site character of the dataset. The scans were acquired on Philips iCT 256, GE LightSpeed16 and Toshiba Aquilion scanners, the last of which appears only in the test set. Each slice is 512×512 pixels, and the number of slices per scan ranges from 157 to 1125, with a slice thickness between 0.45 and 1.00 mm and an in-plane resolution between 0.500 and 0.919 mm. The health status of the subjects ranges from healthy volunteers to patients with severe pulmonary disease. The multi-domain character comes from the 58 COVID-19 scans included in the test set: ground-glass opacity, consolidation and septal thickening alter the appearance of the lung parenchyma and blur the airway lumen, so these cases form a noisy domain on which the generalization ability of a model trained on clean scans can be assessed.

The annotations were produced in two stages. A preliminary segmentation was first obtained by ensembling, through majority voting, several airway segmentation models trained on the BAS dataset.

These preliminary masks were then reviewed by three radiologists with more than five years of experience: two of them independently refined and cross-checked each scan, and the most senior radiologist validated the final annotation. The manual refinement required 60 to 90 minutes per scan and focused mainly on repairing breakages in the peripheral bronchi and removing small leakages into the surrounding parenchyma, using the original CT intensities and anatomical prior knowledge. The whole collection and annotation process took the organizers approximately one year.

The challenge evaluates submissions along two complementary axes. Topological completeness is measured by the tree length detected rate (TD) and the branch detected rate (BD),

$$TD = \frac{T_{det}}{T_{ref}}, \quad BD = \frac{B_{det}}{B_{ref}}, \quad (1)$$

where T_{det} and B_{det} denote respectively the total length and the number of branches correctly detected in the prediction, and T_{ref} and B_{ref} the corresponding quantities in the ground truth. Both metrics are computed on the largest connected component of the prediction, a branch being counted as detected when more than 80% of its centerline voxels lie within the ground truth. Topological correctness is measured by the Dice similarity coefficient (DSC) and the precision, with sensitivity and specificity reported as complementary indicators. The official ranking is based on the unweighted mean of TD, BD, DSC and precision. This dual evaluation reflects the fact that overlap-based measures alone are insufficient for tubular structures: a prediction may achieve a high Dice score while missing a substantial number of distal branches, since the peripheral bronchi account for only a small fraction of the total airway volume.

II The AIB23 dataset

The AIB23 dataset (Airway-Informed Quantitative CT Imaging Biomarker for Fibrotic Lung Disease 2023) was released as part of an official challenge held in conjunction with MICCAI 2023 and hosted on the CodaLab platform (Nan et al., 2024). It is the first publicly available dataset dedicated to airway segmentation in patients with fibrotic lung disease. Earlier public airway datasets, such as EXACT'09, BAS and ATM'22, mainly contain scans from healthy subjects or from patients with common pulmonary conditions such as nodules or COVID-19.

Fibrotic lung disease is particularly challenging for automatic airway segmentation for two reasons. First, honeycombing, the presence of small cystic airspaces in the lung periphery, visually resembles distal airway branches, so models often produce false-positive predictions known as airway leakages. Second, traction bronchiectasis and the distortion of lung tissue alter airway morphology, which frequently leads to missing or discontinuous branches. AIB23 was therefore designed to benchmark segmentation methods on these difficult cases.

Data sources and composition. The dataset gathers high-resolution computed tomography (HRCT) scans from three cohorts, acquired with different vendors and scanning protocols. The training and validation sets come from the Open Source Imaging Consortium (OSIC), an open-access database of patients with interstitial lung disease (ILD), idiopathic pulmonary fibrosis (IPF) and other respiratory diseases. The test set combines 90 fibrosis cases from the Australian IPF Repository (AIPFR) with 50 COVID-19 cases from the University Hospital of Parma. The COVID-19 cases make it possible to assess generalization to a different pathology. The patient populations are broadly comparable in age, with a median age of 68 years for OSIC, 70 for AIPFR and 65 for Parma. Table 2 summarizes the data splits.

Table 2: Composition of the AIB23 dataset (Nan et al., 2024).

Split	Source	Task I (segmentation)	Task II (mortality)
Training	OSIC	120	95
Validation	OSIC	52	52
Test	AIPFR + Parma	140 ^a	90

^a90 fibrosis (AIPFR) and 50 COVID-19 (Parma) cases.

Image selection and preprocessing. All scans were converted from DICOM to NIfTI format using SimpleITK. Only scans with more than 120 slices and an in-plane resolution of at least 512×512 pixels were retained. Despite this criterion, the number of slices varies considerably between cases. The fibrosis test scans contain significantly fewer slices than the training data, which results in lower image quality in the coronal and sagittal planes. This represents a domain shift between training and test data.

Annotation protocol. The ground-truth airway masks were produced through a semi-automatic, multi-stage process. Preliminary labels were first generated by a deep learning model trained on the BAS dataset. Three junior radiologists then revised these labels, and two senior radiologists refined them by correcting breakages and mis-annotated voxels. When annotators disagreed, the most experienced radiologist's annotation was retained. Annotation was performed in ITK-SNAP using a drawing tablet. Inter-observer agreement on airway branches was very high (Cohen's $\kappa = 0.937$).

Tasks and evaluation. The challenge defines two tasks. Task I is binary airway segmentation of the 3D HRCT volume. It is evaluated with overlap metrics (intersection over union, IoU, and precision) and with topological metrics that are particularly relevant for thin tubular structures:

- the detected length ratio (DLR) is the fraction of the ground-truth branch length that the prediction recovers;
- the detected branch ratio (DBR) is the fraction of ground-truth branches that are correctly detected, a branch counting as detected when its IoU with the corresponding ground-truth branch is at least 0.8;
- the airway leakage ratio (ALR) quantifies false-positive predictions.

The challenge's overall accuracy score is the mean of IoU, precision, DLR and DBR. For reference, the best-performing challenge submission reached an IoU of 0.905, a DLR of 0.937 and a DBR of 0.905 on the test set.

Task II is a binary classification problem: predicting whether a patient is alive or deceased 63 weeks after the first HRCT scan. The training set contains 59 survivors and 36 deceased patients, and the validation set is balanced (26/26). The test set is strongly imbalanced, with about 87 % of survivors. Performance is measured with accuracy, AUC, sensitivity, specificity and F1-score.

Registration

[Optional short introduction: why a common spatial reference is needed, and the two steps used to obtain it – building a population template, then registering each airway tree to it.]

II Template construction

To obtain a common spatial reference for the pulmonary airway trees, a population-specific three-dimensional template was constructed from airway segmentation volumes using the Advanced Normalization Tools (ANTs) framework and its ANTsPy interface (Tustison et al., 2021). Template construction is a *groupwise registration problem*: a common template and a set of transformations mapping the individual airway trees to it are estimated together. Unlike pairwise registration to an arbitrarily selected subject, groupwise registration does not designate one individual airway tree as the anatomical reference; the reference itself is progressively optimised to represent the population. The ANTs framework implements this as *symmetric groupwise normalization* (SyGN) (Avants et al., 2010). The registration of individual airway trees to the finished template is described separately in Section “Affine registration”.

Sources of the method description. The description combines three sources. (i) The pairwise deformable registration algorithm, Symmetric Normalization (SyN), was introduced by Avants et al. (2008) (Section “Pairwise registration with SyN”). (ii) The groupwise template-construction strategy, SyGN, was introduced by Avants et al. (2010) (Section “Groupwise template construction with SyGN”). (iii) The template initialisation, the appearance update, the parameter defaults and the order of operations are properties of the ANTsPy implementation (`ants.registration` and `ants.build_template`) (Tustison et al., 2021); where the implementation differs from the published algorithms, this is stated explicitly. Both methods were developed and validated on T1-weighted brain MRI (cortical labelling in elderly and neurodegenerative brains, and hippocampus segmentation in epilepsy, respectively); neither was evaluated on airway segmentations.

III Input data and image geometry

Selection of the input volumes. Eight airway segmentation volumes were used: four drawn at random from ATM’22 and four from AIIB23 (Section “Datasets”). The draw used Python’s pseudo-random generator without a fixed seed, so the selected subset is not reproducible from the code alone; the file names were recorded in the run log and are listed in Table 3. The volumes were passed to

Table 3: Airway segmentations used to build the template, in input order.

Order	Dataset	File name
1	ATM’22	[TODO: file name]
2	ATM’22	[TODO: file name]
3	ATM’22	[TODO: file name]
4	ATM’22	[TODO: file name]
5	AIIB23	[TODO: file name]
6	AIIB23	[TODO: file name]
7	AIIB23	[TODO: file name]
8	AIIB23	[TODO: file name]

`ants.build_template` in the order ATM’22 first, then AIIB23. This order matters for one implementation detail: the first volume defines the voxel grid of the template (see *Initialisation* in Section “Groupwise template construction with SyGN”).

Representation in physical space. Each airway tree is a three-dimensional NIfTI image: a discrete voxel array together with spatial metadata that positions the array in physical space. A voxel is identified by a discrete index $\mathbf{i} = (i_x, i_y, i_z)^T$, whereas its anatomical location is a continuous physical coordinate $\mathbf{x} = (x, y, z)^T$, related by

$$\mathbf{x} = \mathbf{o} + \mathbf{D} \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & s_z \end{bmatrix} \mathbf{i}, \quad (2)$$

where $\mathbf{o}$ is the image origin, s_x, s_y, s_z are the voxel spacings and $\mathbf{D}$ is the direction matrix of the voxel axes; in NIfTI terminology these are encoded by the spatial affine in the image header. ANTs is built on ITK, in which registration is performed in *physical coordinates*, and ANTsPy converts the NIfTI geometry into the ITK representation of origin, spacing and direction. Differences in origin, orientation or voxel spacing between two airway segmentations are therefore taken into account when their structures are put into correspondence. The NIfTI affine defines the *initial coordinate system of each image*; it is not itself a registration.

Warping convention. An image I is treated as a function of physical position, and a transformation is applied by composition. If T is the fixed image (the template) and I the moving image, the warped image is defined on the domain of T :

$$(I \circ \Phi)(\mathbf{x}) = I(\Phi(\mathbf{x})), \quad \mathbf{x} \in \Omega_T. \quad (3)$$

Hence Φ maps each point of the *template* domain to its corresponding point in the *subject* domain, where the subject image is resampled. This is the convention used by ITK and ANTs; the inverse map Φ^{-1} takes subject points to template points.

Pairwise registration with SyN

Within every template-building iteration k , each airway segmentation I_i is registered to the current template $T^{(k)}$, which serves as the fixed image. The registration seeks $\Phi_i^{(k)}$ such that

$$I_i \circ \Phi_i^{(k)} \approx T^{(k)}, \quad (4)$$

i.e. after warping, corresponding airway structures occupy similar physical locations in template space. Inside `ants.build_template` this is the call

```
ants.registration(
    fixed=xavg, # current template T^(k)
    moving=image_i, # airway segmentation I_i
    type_of_transform='SyN'
)
```

The ANTsPy SyN preset. Despite its name, the ANTsPy preset SyN contains three stages. The two images are first aligned by their centres of mass (a translation used as initial transform), an affine transformation $A_i^{(k)}$ is then estimated, and finally the deformable symmetric normalisation $\phi_i^{(k)}$ is estimated between the template and the affinely transformed subject $I_i \circ A_i^{(k)}$. In Avants et al. (2008), by contrast, “SyN” denotes only the deformable part, and rigid and scaling differences are assumed to have been removed beforehand. With the convention of Eq. (3), the total transformation is

$$\Phi_i^{(k)} = A_i^{(k)} \circ \phi_i^{(k)}, \quad I_i \circ \Phi_i^{(k)} = (I_i \circ A_i^{(k)}) \circ \phi_i^{(k)}, \quad (5)$$

so a template point is moved first by the deformable map and then by the affine map into subject space. The affine part accounts for global translation, rotation, scaling and shear; the deformable part accounts for spatially varying differences that no single affine map can represent. (The centre-of-mass translation is folded into the affine stage.)

Registration parameters. Apart from `type_of_transform`, no argument was specified, so all other settings were ANTsPy defaults: Mattes mutual information as the similarity metric of both the affine and the deformable stage; an affine stage with (2100, 1200, 1200, 10) iterations on four resolution levels; a deformable stage with (40, 20, 0) iterations from coarse to fine; an optimiser step `grad_step` = 0.2; and Gaussian smoothing of the update field (`flow_sigma` = 3) and of the total field (`total_sigma` = 0). Because the last entry of the deformable iterations is zero, no deformable optimisation is performed at full image resolution, which limits how precisely thin peripheral airway branches can be aligned. The SyN step `grad_step` is unrelated to the template shape-update step `gradient_step` of Section “Groupwise template construction with SyGN”, although both equal 0.2.

Similarity metric. Mutual information (MI) measures the statistical dependence between the intensity distributions of the fixed and warped moving image, estimated from their joint histogram. It is *not* the metric used in either reference publication. Avants et al. (2008) developed SyN with *local cross-correlation* (CC), arguing that MI is a global measure well suited to rigid registration but potentially limiting for deformable registration, and that a localised MI estimate becomes unreliable because joint probabilities need many samples; local CC depends only on local means and variances, which can be estimated from few samples. SyGN (Avants et al., 2010) likewise uses local CC both for the registrations and for the template appearance. With the mean-subtracted images $\bar{I} = I - \mu_I$ and $\bar{J} = J - \mu_J$, the means being computed in a small cubic neighbourhood of $\mathbf{x}$, local CC is

$$CC(\bar{I}, \bar{J}, \mathbf{x}) = \frac{\langle \bar{I}, \bar{J} \rangle^2}{\langle \bar{I}, \bar{I} \rangle \langle \bar{J}, \bar{J} \rangle}, \quad (6)$$

with inner products over the same neighbourhood (5^3 voxels in Avants et al. (2008), radius 4 in Avants et al. (2010)). Because the airway volumes are segmentations, any metric compares the spatial patterns of the label values rather than anatomical intensities.

Registration as optimisation. Deformable registration is often written as

$$\hat{\Phi} = \arg \min_{\Phi} D(T, I \circ \Phi) + \lambda R(\Phi), \quad (7)$$

with a dissimilarity term D , a regulariser R and a weight λ . This is the classical elastic formulation. SyN departs from it: Avants et al. (2008) restrict the search to diffeomorphisms and optimise the similarity term essentially without a competing penalty. Regularity comes from the diffeomorphic path structure and from smoothing of the velocity field; the regularisation term contributes little to the total energy but ensures that a path of minimal length is found.

Diffeomorphisms generated by velocity fields. A diffeomorphism is a differentiable map with a differentiable inverse. In SyN, as in other large-deformation methods, it is generated by integrating a time-dependent velocity field $\mathbf{v}(\mathbf{x}, t)$:

$$\frac{\partial \phi(\mathbf{x}, t)}{\partial t} = \mathbf{v}(\phi(\mathbf{x}, t), t), \quad \phi(\mathbf{x}, 0) = \mathbf{x}. \quad (8)$$

The regularity of $\mathbf{v}$ is measured by a norm $\|\mathbf{v}\|_L$ induced by a linear differential operator such as $L = a\nabla^2 + b\text{Id}$, and the path length

$$D(\phi(\cdot, 0), \phi(\cdot, 1)) = \int_0^1 \|\mathbf{v}(\cdot, t)\|_L dt \quad (9)$$

defines a distance between shapes; its infimum over admissible velocity fields is a true metric, whose shortest paths are geodesics. This distance is what SyGN later minimises to find the average shape (Section “Groupwise template construction with SyGN”).

Symmetric two-part formulation. SyN splits the path between two images I and J into two halves. Two diffeomorphisms, ϕ_1 acting on I and ϕ_2 acting on J , are generated by Eq. (8) with their own velocity fields, each integrated from the identity only over $t \in [0, 0.5]$. The two images are deformed towards each other and compared at the midpoint (figure 1):

$$E_S(I, J) = \inf_{\phi_1} \inf_{\phi_2} \int_0^{0.5} \left(\|\mathbf{v}_1(\cdot, t)\|_L^2 + \|\mathbf{v}_2(\cdot, t)\|_L^2 \right) dt + \int_{\Omega} \Pi(I \circ \phi_1(\cdot, 0.5), J \circ \phi_2(\cdot, 0.5)) d\Omega, \quad (10)$$

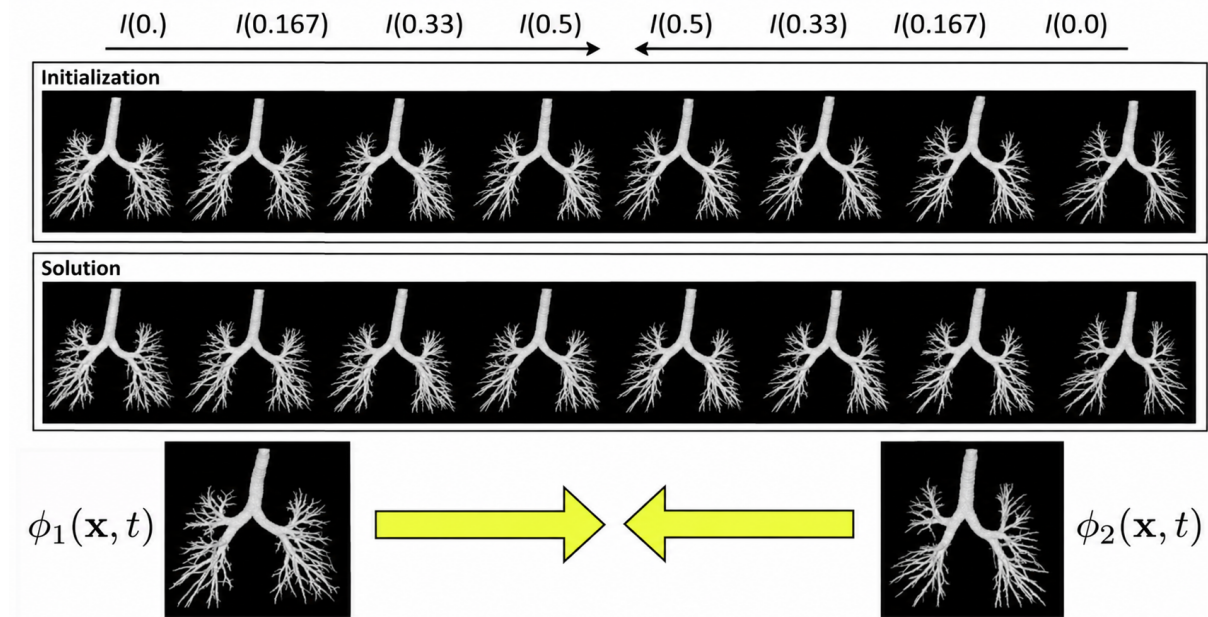


Figure 1: Schematic of the symmetric SyN path between two airway trees I (left) and J (right). Top: the two images at initialisation. Middle: after convergence, both images are deformed along ϕ_1 and ϕ_2 and meet at the mean shape at $t = 0.5$. The full map from I to J is obtained by joining the two half-paths (Eq. 11). Illustrative only; not computed from the study data.

where Π is a pointwise dissimilarity (the squared intensity difference in the derivation of Avants et al. (2008), $-CC$ in their main method). The two half-paths are kept at equal length, so both images travel the same distance and meet at the *mean shape* between them; SyN thus solves a two-image mean problem, the pairwise counterpart of the template estimation in Section “Groupwise template construction with SyGN”.

Full transformations by composition. The complete mapping between the two images is obtained by joining the half-paths:

$$\phi_1(\mathbf{x}, 1) = \phi_2^{-1}(\phi_1(\mathbf{x}, 0.5), 0.5), \quad \phi_1^{-1}(\mathbf{x}, 1) = \phi_1^{-1}(\phi_2(\mathbf{x}, 0.5), 0.5). \quad (11)$$

Because I and J enter Eq. (10) in the same way, exchanging them yields the same path in the opposite direction: the result does not depend on which image is called “fixed” and which “moving”.

Explicit, accurate inverses. Diffeomorphisms are invertible in the continuous setting, but interpolation errors in a discrete implementation accumulate and can break invertibility. SyN therefore computes the inverses ϕ_i^{-1} explicitly with a fixed-point algorithm and enforces $\phi_i^{-1}(\phi_i) = \text{Id}$ and $\phi_i(\phi_i^{-1}) = \text{Id}$ to sub-voxel accuracy. This distinguishes SyN from inverse-consistent registration, in which forward and backward maps are estimated separately and made consistent only approximately through penalty terms. Both the subject-to-template and the template-to-subject mappings are therefore available and consistent.

Optimisation in practice. Each SyN iteration deforms both images to the midpoint, computes the gradient of the similarity term with respect to each half-map, smooths it into velocity updates, updates ϕ_1 and ϕ_2 through a discretisation of Eq. (8) with a sub-voxel step, recomputes the inverses and forms the maps of Eq. (11). The operator L is approximated by a Gaussian kernel applied to the *incremental* update rather than to the total displacement (`flow_sigma` and `total_sigma` in ANTsPy); smoothing the increment is what permits large diffeomorphic deformations, whereas smoothing the total displacement, as in elastic and Demons-type methods, restricts the solution to small deformations. The optimisation runs from coarse to fine resolution levels.

III Groupwise template construction with SyGN

Formulation. SyGN seeks the template and the set of transformations that give the “smallest” parameterisation of the dataset, measured by the SyN energy of Eq. (10). The template is described by an appearance $\bar{I}$ (its intensities) and a shape, represented by a diffeomorphism ψ that defines the template’s coordinate system. Given the images $\{J^i\}_{i=1}^N$, Avants et al. (2010) minimise

$$E_{\bar{I}} = \sum_{i=1}^N E_S(\bar{I}, J^i, \phi^i) \quad \text{with} \quad \phi^i(\mathbf{x}, 0) = \psi(\mathbf{x}) \quad \forall i, \quad (12)$$

jointly over the mappings ϕ^i , the appearance $\bar{I}$ and the shape ψ . No initial template has to be chosen by the user, and the stated goals are that the final template’s appearance does not depend on any specific anatomy and its shape does not depend on any individual’s coordinate system.

Coordinate-descent algorithm. Eq. (12) is minimised by coordinate descent, one group of variables at a time with the others fixed. After initialisation, each iteration (1) registers every image to the fixed template with SyN; (2) updates the template appearance $\bar{I}$; (3) updates the template shape ψ towards the average of the mappings; and (4) resamples the template through the new ψ , resets the mappings to start from it, and repeats. The template is only updated after all registrations of an iteration are complete, so appearance and shape are never estimated from partially optimised mappings. `ants.build_template` implements the same loop, but with a different initialisation and a different appearance update, as described next (figure 2).

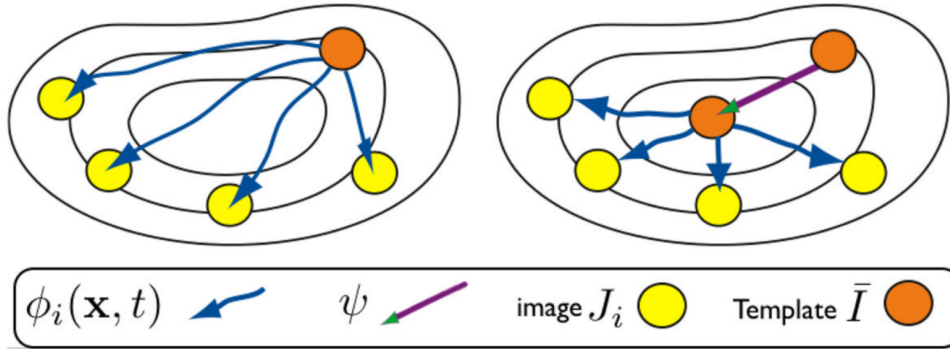


Figure 2: Two steps of one SyGN iteration. Left: the diffeomorphic paths ϕ_i from the current template $\bar{I}$ (orange) to the images J_i (yellow) are estimated (Step 1). Right: the template shape is moved by ψ towards the population centre, which shortens the total path length (Step 3). Reproduced from Avants et al. (2010), Fig. 2.

Initialisation. In Avants et al. (2010), the first template is the voxelwise (Euclidean) average of the images *after affine alignment*, and all ϕ^i and ψ start as the identity. In the present implementation no initial template was supplied, and ANTsPy forms the initial template *without any alignment*: each image is multiplied by its weight $w_i = 1/N$, resampled onto the voxel grid of the first input image I_1 , and summed,

$$T^{(0)}(\mathbf{x}) = \sum_{i=1}^N w_i I_i(\mathbf{x}), \quad \mathbf{x} \in \Omega_{I_1}. \quad (13)$$

Two consequences follow. First, $T^{(0)}$ averages the airway trees as they lie in their own scanner coordinates, so it is blurred and its quality depends on how well these coordinate systems already agree. Because the inputs are binary segmentations, it also contains intermediate values equal to the fraction of volumes that contain airway at each location. Second, every later template is resampled on this grid, so the origin, spacing, orientation and field of view of the final template are inherited from I_1 , here the first randomly selected ATM'22 volume. The template's *shape* is still optimised over the whole population, but its *grid* is not independent of the input order.

Step 1 – registration. With $T^{(k)}$ fixed, each I_i is registered to it with the ANTsPy SyN preset (Section “Pairwise registration with SyN”), giving the affine maps $A_i^{(k)}$, the deformable maps $\phi_i^{(k)}$ and the warped images $I_i \circ \Phi_i^{(k)}$, all on the template grid.

Step 2 – appearance update. Avants et al. (2010) do not take the template appearance to be a plain average. They optimise it by gradient ascent on the cross-correlation of Eq. (6) between the template and the warped images (intensities normalised to $[0, 1]$, gradients smoothed with a Gaussian of $\sigma = 1$ voxel, step size 0.1), and show that this preserves more contrast and detail than the Euclidean mean. ANTsPy does *not* implement this step. Its appearance is the weighted Euclidean mean of the warped images,

$$T_{\text{avg}}^{(k+1)}(\mathbf{x}) = \sum_{i=1}^N w_i \left[I_i \circ \Phi_i^{(k)} \right](\mathbf{x}), \quad (14)$$

and, at the end of each iteration (after the shape update below), this image is blended with a sharpened copy of itself,

$$T^{(k+1)} = \beta T_{\text{shape}}^{(k+1)} + (1 - \beta) \text{Sharpen} \left(T_{\text{shape}}^{(k+1)} \right), \quad (15)$$

with $\beta = \text{blending_weight} = 0.75$. The sharpening blend is a heuristic of the implementation that compensates for the blurring of the mean; it appears in neither publication and does not deform the template. In both the paper and the implementation, averaging happens *after* spatial alignment, so each template location combines corresponding locations of the airway trees rather than identical voxel indices of the original images.

Step 3 – shape update. Averaging aligned images fixes the appearance but not the *position* of the template within the population: if the current template is closer to some subjects than to others, the mappings to those subjects are short and the others long (figure 2, right). Avants et al. (2010) therefore move the template towards the Fréchet (intrinsic) mean of the diffeomorphisms, the shape ψ that minimises the total squared geodesic distance $\sum_i D^2(\psi, \phi^i(\cdot, 1))$ of Eq. (9). Velocity fields, unlike diffeomorphisms, can be added, so the average can be taken over the initial velocity fields of the mappings; setting the derivative of the total distance to zero gives the optimality condition

$$\bar{\mathbf{v}} = \sum_{i=1}^N \mathbf{v}^i(\psi, 0) = 0, \quad (16)$$

and, away from the optimum, the template shape evolves along $d\psi/d\bar{t} = \bar{\mathbf{v}}(\psi, \bar{t})$. Under a small-deformation approximation, with $\psi = \text{Id}$ at the current template, one step of length $\Delta\bar{t}$ becomes a scaled sum of the displacements of the mappings:

$$\psi_{\Delta\bar{t}}(\mathbf{x}) = \mathbf{x} + \frac{\Delta\bar{t}}{\Delta t} \sum_{i=1}^N (\phi_1^i(\mathbf{x}, \Delta t) - \mathbf{x}). \quad (17)$$

At the average configuration the displacements cancel and the template no longer moves. ANTsPy implements this approximation: it averages the forward displacement fields $W_i^{(k)}$ of the deformable registrations, $W^{(k)} = \sum_i w_i W_i^{(k)}$, scales the result by $-\alpha$ with $\alpha = \text{gradient_step} = 0.2$ (the counterpart of $\Delta\bar{t}$), combines it with the inverse of the average affine transformation $\bar{A}^{(k)}$ of the registrations, and resamples the averaged image through the resulting map $\Psi^{(k)}$:

$$T_{\text{shape}}^{(k+1)} = T_{\text{avg}}^{(k+1)} \circ \Psi^{(k)}, \quad \Psi^{(k)} \text{ built from } -\alpha W^{(k)} \text{ and } (\bar{A}^{(k)})^{-1}. \quad (18)$$

Two simplifications relative to the paper should be noted: ANTsPy averages total displacement fields rather than initial velocity fields, and it also re-centres the template affinely. With `useNoRigid=True` (the default), the rigid part is excluded from $\bar{A}^{(k)}$, so only the average scaling and shear are removed and the template does not drift by an arbitrary common rotation or translation. Eq. (18) is schematic; the exact composition order follows the ANTsPy source.

Why the shape update matters for segmentation volumes. Avants et al. (2010) illustrate this step (Figure 3) with a synthetic set of four binary ellipses with the same centre and different radii. The initial average has intermediate grey levels, like the averaged airway masks here. With the shape update, SyGN converges, up to interpolation error, to the true mean shape. Without it, optimising appearance alone converges to a geometrically wrong template, the 0.5 level set of the initial average. The shape update is therefore what makes the geometry of a template built from binary segmentations reflect the population average rather than the initialisation; setting `gradient_step > 0` is essential for this.

Iterations and convergence. The loop was run for a fixed number of 8 iterations (the ANTsPy default is 3). Avants et al. (2010) report that SyGN typically converges in fewer than ten iterations, usually three

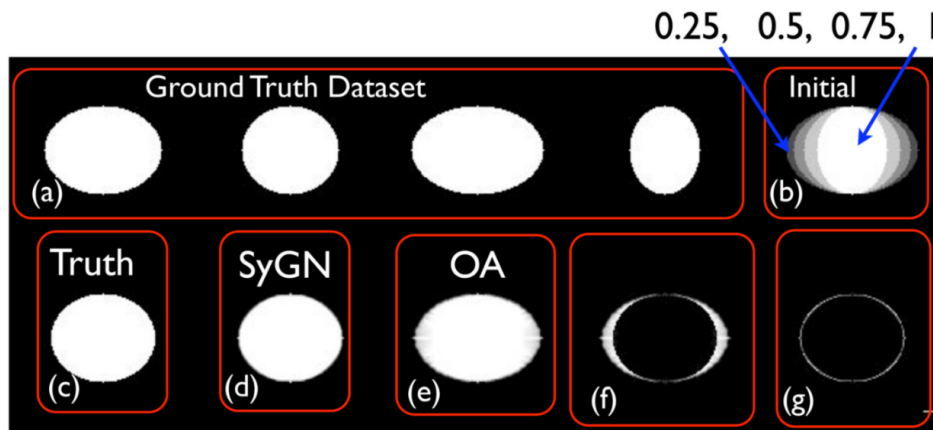


Figure 3: Effect of the shape update on binary images. (a) Four binary ellipses; (b) their initial average, with grey levels 0.25–1; (c) true mean shape; (d) SyGN result with shape update; (e) result without shape update (appearance only), which converges to the wrong shape; (f,g) errors of (e) and (d) with respect to (c). Reproduced from Avants et al. (2010), Fig. 3.

to five, depending on the complexity of the deformations. No stopping criterion is applied by the code. However, ANTsPy prints the mean absolute value of the average displacement field $W^{(k)}$ at every iteration; by Eq. (16) this quantity should tend towards zero as the template approaches the population mean, and it can be used to assess convergence.

Unbiasedness and its limits. In principle SyGN gives a template that is unbiased: it requires no user-selected reference, uses a symmetric pairwise method, and centres the template shape on the population. In the present implementation this holds for the template shape up to three qualifications: the template grid is inherited from the first input volume (Eq. (13)); the eight volumes are a small random subset of the two datasets; and the registrations and the appearance use MI and a sharpened mean instead of the cross-correlation model of the reference publications.

Output. After the eighth iteration, the template $T^{(8)}$ was written to `template.nii.gz`. It is the reference used in Section “Affine registration”. The complete procedure is summarised in Figure 4.

The NIfTI geometry defines where each original voxel grid lies in physical space, whereas the affine and diffeomorphic transformations estimated by ANTs establish anatomical correspondence between these physical domains. The final template is therefore a population-derived coordinate space in which the anatomical variability of the eight airway trees has been brought into correspondence.

II Affine registration

III Claude

Objective. Airway trees extracted from different subjects differ in position, orientation and overall size, as a consequence of patient positioning in the scanner, differences in body size and acquisition parameters. To make them directly comparable, all airway segmentations were mapped into a common spatial reference frame defined by a template airway tree. An affine transformation was chosen for this purpose because it compensates for these global differences while preserving the local geometry of

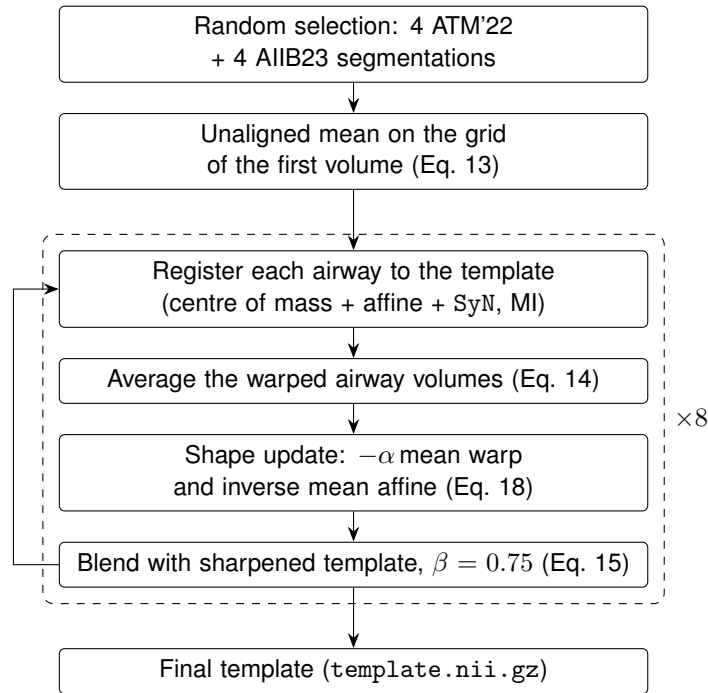


Figure 4: Groupwise template construction with `ants.build_template`. The dashed block is one SyGN iteration, repeated eight times.

each airway tree, which is the object of the subsequent analysis (Maintz and Viergever, 1998).

Data. The input data consisted of the **[TODO: N]** ground-truth airway segmentations of the AIB23 challenge dataset (Nan et al., 2024), stored as binary three-dimensional volumes in NIfTI format. A single airway tree (template 22) was used as the reference image for all registrations.

Transformation model. Registration consists of finding the spatial transformation T that best aligns a *moving* image M with a *fixed* image F . In this work, the fixed image is the template and each AIB23 segmentation is, in turn, the moving image. The transformation is restricted to the class of three-dimensional affine maps,

$$T_{\mu}(\mathbf{x}) = \mathbf{A}(\mathbf{x} - \mathbf{c}) + \mathbf{c} + \mathbf{t}, \quad (19)$$

where $\mathbf{x} \in \mathbb{R}^3$ is a point in the fixed image domain, $\mathbf{A} \in \mathbb{R}^{3 \times 3}$ is a general matrix, $\mathbf{t} \in \mathbb{R}^3$ is a translation vector and $\mathbf{c}$ is a fixed centre of rotation. The parameter vector μ therefore contains $9 + 3 = 12$ degrees of freedom, which jointly encode translation, rotation, anisotropic scaling and shear. Because the transformation is linear up to a translation, straight lines are mapped to straight lines and parallelism is preserved, so the shape of the airway tree is modified only globally.

Similarity metric and optimisation. The optimal parameters are obtained by solving

$$\hat{\mu} = \arg \min_{\mu} \mathcal{C}(F, M \circ T_{\mu}), \quad (20)$$

where $\mathcal{C}$ is a cost function measuring the dissimilarity between the fixed image and the transformed moving image. The cost function used is the negative of the Mattes mutual information (Mattes et al.,

2003),

$$\mathcal{C}(\boldsymbol{\mu}) = - \sum_{\ell} \sum_{\kappa} p(\ell, \kappa; \boldsymbol{\mu}) \log \frac{p(\ell, \kappa; \boldsymbol{\mu})}{p_F(\ell) p_M(\kappa; \boldsymbol{\mu})}, \quad (21)$$

where $p(\ell, \kappa; \boldsymbol{\mu})$ is the joint probability of intensity bins ℓ and κ in the fixed and transformed moving images, and p_F and p_M are the corresponding marginal distributions. In the Mattes formulation, these distributions are estimated with B-spline Parzen windows, which makes the metric differentiable with respect to $\boldsymbol{\mu}$. The histograms were built with 32 bins from a random sample of 20% of the voxels, which reduces the computational cost of each iteration.

Equation (20) was solved with a gradient descent optimiser. Before optimisation, the transformation was initialised by aligning the intensity centres of mass of the two images, which provides a coarse translational alignment and reduces the risk of convergence to a poor local optimum.

Multi-resolution strategy. To improve robustness and speed, the optimisation was performed within a four-level multi-resolution scheme (Thévenaz et al., 1998). At each level, both images were smoothed with a Gaussian kernel and downsampled by an integer shrink factor; the optimisation then proceeded from the coarsest level to the original resolution, with the result of each level used to initialise the next. Coarse levels capture the large-scale alignment, while finer levels refine it. The settings of each level are listed in Table 4.

Table 4: Multi-resolution settings of the affine registration.

Level	Shrink factor	Gaussian σ (voxels)	Maximum iterations
1 (coarsest)	6	3	2100
2	4	2	1200
3	2	1	1200
4 (finest)	1	0	10

At each level, the optimisation stopped either when the maximum number of iterations was reached or when the cost function had converged, i.e. when its variation over the last 10 iterations fell below 10^{-6} .

Resampling of the registered images. Once the optimal transformation $T_{\hat{\boldsymbol{\mu}}}$ had been estimated, each moving segmentation was resampled onto the voxel grid of the template using linear interpolation and saved in NIfTI format. Since the inputs are binary masks, linear interpolation yields intermediate values in $[0, 1]$ at the boundary of the airway lumen, while voxels well inside or outside the airway keep the values 1 and 0 respectively.

Implementation. The registration was implemented in Python 3.11 with ANTsPy (version [TODO: x.y.z]) (Tustison et al., 2021), the Python interface of the Advanced Normalization Tools (ANTs) (Avants et al., 2009), which is built on the registration framework of the Insight Toolkit (ITK) (Avants et al., 2014). The affine registration was obtained with the function `ants.registration` using the predefined Affine transformation type, whose default settings correspond to the parameters described above under *Similarity metric and optimisation* and *Multi-resolution strategy*.

Since each segmentation is registered to the template independently of the others, the task is embarrassingly parallel. The computation was run on a single node of a high-performance computing cluster managed by the SLURM workload manager (Yoo et al., 2003), with 48 CPU cores and 128 GB of memory allocated to the job. Parallelism was organised on two levels: the joblib library (Joblib Development

Team, n.d.) distributed the registrations over 12 independent worker processes, and each registration used 4 ITK threads, so that

$$12 \text{ processes} \times 4 \text{ threads} = 48 \text{ cores}, \quad (22)$$

matching the allocated resources exactly. Segmentations for which a registered output already existed were skipped, which allowed an interrupted job to be resumed without repeating completed registrations.

Because the similarity metric is estimated from a random subset of voxels and no fixed random seed was set, repeating a registration may yield slightly different transformation parameters; these differences are expected to be small relative to the global misalignment corrected by the affine transformation.

III ChatGPT

Following template construction, the airway segmentation volumes were spatially aligned to a common anatomical reference using affine image registration. The registration was performed using the Advanced Normalization Tools (ANTs) framework through its Python interface, ANTsPy. ANTs is an open-source framework for quantitative medical image analysis whose core functionality includes image registration and template construction (Avants et al., 2009; Tustison et al., 2021). For each subject, the corresponding three-dimensional airway segmentation volume from the AIB23 dataset was considered as the *moving* image, whereas the previously constructed airway template, `template_22.nii.gz`, was used as the *fixed* image. In the implemented pipeline, all available input volumes were processed independently and registered to this fixed reference.

Affine transformation. The affine registration models global geometric differences between an individual airway tree and the reference template. In three-dimensional space, an affine transformation can be written as

$$\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{t}, \quad (23)$$

where $\mathbf{x} \in \mathbb{R}^3$ denotes a point in the original image space, $\mathbf{x}' \in \mathbb{R}^3$ its transformed position, $\mathbf{A} \in \mathbb{R}^{3 \times 3}$ is the linear component of the transformation, and $\mathbf{t} \in \mathbb{R}^3$ is the translation vector. The matrix $\mathbf{A}$ can represent rotation, scaling, and shearing, whereas $\mathbf{t}$ accounts for translation (Johnson et al., 2015). Consequently, a three-dimensional affine transformation contains twelve parameters, corresponding to the nine elements of the linear transformation matrix and the three translation parameters. The ANTsPy documentation explicitly defines its *Affine* registration option as an affine transformation with 12 parameters (ANTsPy Developers, n.d.).

In contrast to deformable registration, the affine transformation is global: the same transformation is applied throughout the image. It can therefore correct subject-specific differences in overall position, orientation, scale, and global shear, but it does not introduce spatially varying local deformations. Consequently, the relative geometric configuration of the airway tree is preserved while the tree is brought into a common global coordinate system (Johnson et al., 2015; Avants et al., 2009).

Registration as an optimization problem. Let F denote the fixed template and M the moving subject image. The registration seeks the affine transformation parameters θ that optimize a similarity-based registration objective between the fixed image and the transformed moving image. In general, the registration problem can be expressed as

$$\theta^* = \arg \underset{\theta}{\text{opt}} \mathcal{C}(F, M \circ T_{\theta}), \quad (24)$$

where T_θ is the affine transformation parameterized by θ , $M \circ T_\theta$ denotes the moving image after transformation, and $\mathcal{C}$ denotes the registration objective. Image registration frameworks such as ITK formulate registration through the combination of a transformation model, a similarity metric, and an optimization procedure (Johnson et al., 2015; Avants et al., 2014).

In the present implementation, the registration is invoked through the ANTsPy interface as

```
ants.registration(
    fixed=fixed,
    moving=moving,
    type_of_transform='Affine'
)
```

where the fixed image is the image defining the reference space and the moving image is the image mapped to that space (ANTsPy Developers, n.d.). The choice `type_of_transform='Affine'` therefore specifies that the estimated transformation belongs to the affine transformation family rather than to a deformable model.

The implementation does not explicitly specify additional registration parameters such as the similarity metric, optimization schedule, number of iterations, or multiresolution settings. These are therefore determined by the ANTsPy Affine registration preset associated with the software version used in the experiment. The ANTsPy documentation exposes these parameters explicitly, including the affine metric, sampling strategy, iteration schedule, shrink factors, and smoothing sigmas (ANTsPy Developers, n.d.). This distinction is important for reproducibility: only parameters explicitly specified in the experimental code should be reported as manually selected parameters.

Mapping and resampling to template space. After estimating the affine transformation, the transformed moving image is resampled onto the spatial domain of the fixed image. In the ITK registration framework, the moving image is evaluated at the physical coordinates induced by the estimated transformation, and an interpolator is used when mapped locations do not coincide with the original voxel grid (Johnson et al., 2015). The purpose of this resampling step is to obtain a representation of the moving image on the grid of the fixed image, allowing the registered volumes to share a common spatial reference.

In the implemented pipeline, the registered volume is obtained from the `warpedmovout` output returned by `ants.registration`. According to the ANTsPy documentation, `warpedmovout` is the moving image warped into the space of the fixed image (ANTsPy Developers, n.d.). The resulting volume is subsequently written as a new NIfTI file, thereby producing an airway segmentation expressed in the coordinate system of the template.

The complete procedure can therefore be summarized as

$$\begin{array}{ccc} \text{subject airway segmentation} & \xrightarrow{\text{affine registration}} & \text{estimated affine transformation} \\ & \searrow \text{resampling} & \\ & & \text{registered segmentation in template space.} \end{array} \quad (25)$$

The resulting registered airway volumes share a common spatial reference and can consequently be used for subsequent inter-subject analysis, including morphological comparison, shape analysis, and representation learning.

Implementation and computational setup. The registration was applied independently to all available AIB23 airway segmentation volumes. The implementation uses a parallelized execution strategy,

with 12 registration jobs and 4 ITK threads assigned to each job. These settings concern the computational execution of the pipeline and do not change the mathematical definition of the affine transformation itself.

The registered volumes were saved in the specified output directory using the subject identifier followed by the suffix `_R.nii.gz`. These files represent the final airway segmentations after affine alignment to the common template and were used as inputs to the subsequent processing stages.

I Encoding

[Describe how the data is represented or encoded (features, embeddings, formats), why you chose this representation, and what it captures.]

I Labelling

[Explain the labelling process: the classes or categories, the tools used, who labelled the data, and how label quality or consistency was checked.]

I Morphological Metrics

[List the morphological metrics you computed, define each one (a formula can help), explain how it is calculated and why it is relevant.]

I Clustering

[Describe the clustering method, its inputs and parameters (e.g. number of clusters), how the result was evaluated, and what the clusters show.]

I Conclusions and Recommendations

II Conclusions

[Summarise how the objectives stated in the introduction have been met and make some concluding remarks.]

II Recommendations

[Briefly describe the proposed plans of action for the future.]



I Appendices

II Appendix A: Title of the Document

[Any document or information needed to support an argument or clarify something mentioned in the body of the report. Number and title every table and figure.]



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